D 36. Wykaż, że $\lim\limits_{n\to\infty}\left(\frac{1}{\sqrt{2}}-\frac{1}{2}+\frac{1}{2\sqrt{2}}-\frac{1}{4}+\ldots+\frac{1}{2^{n-1}\cdot\sqrt{2}}-\frac{1}{2^n}\right)=\sqrt{2}-1.$

$$\frac{1}{\sqrt{2}}-\frac{1}{2}+\frac{1}{2\sqrt{2}}-\frac{1}{4}+\ldots+\frac{1}{2^{n-1}\cdot\sqrt{2}}-\frac{1}{2^n}$$

$$a_1=\frac{1}{\sqrt{2}} \qquad a_2=-\frac{1}{2} \qquad q=\frac{-\frac{1}{2}}{\frac{1}{\sqrt{2}}}=-\frac{1}{2}\cdot\frac{\sqrt{2}}{1}=-\frac{\sqrt{2}}{2}$$

$$S=\frac{1}{\sqrt{2}}\cdot\frac{1-\left(-\frac{\sqrt{2}}{2}\right)^n}{1+\frac{\sqrt{2}}{2}}=\frac{1}{\sqrt{2}}\cdot\frac{1-\left(-\frac{\sqrt{2}}{2}\right)^n}{\frac{2+\sqrt{2}}{2}}=\frac{1}{\sqrt{2}}\cdot\left(\frac{2}{2+\sqrt{2}}\right)\left(1-\left(-\frac{\sqrt{2}}{2}\right)^n\right)=$$

$$=\left(\frac{2}{2\sqrt{2}+2}\right)\left(1-\left(-\frac{\sqrt{2}}{2}\right)^n\right)=\left(\frac{1}{\sqrt{2}+1}\right)\left(1-\left(\frac{\sqrt{2}}{2}\right)^n\right)=$$

$$=\left(\frac{\sqrt{2}-1}{(\sqrt{2}+1)(\sqrt{2}-1)}\right)\left(1-\left(\frac{\sqrt{2}}{2}\right)^n\right)=(\sqrt{2}-1)\left(1-\left(-\frac{\sqrt{2}}{2}\right)^n\right)$$

$2-1$

$$\lim_{n\to\infty}(\sqrt{2}-1)\left(1-\left(-\frac{\sqrt{2}}{2}\right)^n\right)=\left[(\sqrt{2}-1)(1-0)\right]=\underline{\underline{\sqrt{2}-1}}$$ ▨

37. Rozwiąż nierówność $x^3-\frac{1}{2}x^4+\frac{1}{4}x^5-\frac{1}{8}x^6+\ldots \leqslant \lim\limits_{n\to\infty}\frac{2n^3+6n+5}{(1-n)(2-3n)n}$.

$$x^3-\frac{1}{2}x^4+\frac{1}{4}x^5-\frac{1}{8}x^6 \leqslant \lim_{n\to\infty}\frac{2n^3+6n+5}{(1-n)(2-3n)n}$$

$$a_1=x^3 \qquad a_2=-\frac{1}{2}x^4$$

$$q=\frac{-\frac{1}{2}x^4}{x^3}=-\frac{1}{2}x$$

$$\frac{2n^3+6n+5}{(1-n)(2-3n)n}=\frac{2n^3+6n+5}{2n-3n^2-2n^2+3n^3}=\frac{2n^3+6n+5}{3n^2-5n^2+2n}$$

$$\lim_{n\to\infty}\frac{2n^3+6n+5}{3n^3-5n^2+2n}=\lim_{n\to\infty}\frac{\frac{2n^3}{n^3}+\frac{6n}{n^3}+\frac{5}{n^3}}{\frac{3n^3}{n^3}-\frac{5n^2}{n^3}+\frac{2n}{n^3}}=\frac{2}{3}$$

$$\left|-\frac{1}{2}x\right|<1$$

$$x\in(-2,2)$$

$$S=\frac{x^3}{1+\frac{1}{2}x}=\frac{x^3}{\frac{2+x}{2}}=$$

$$=x^3\cdot\frac{2}{2+x}=\frac{2x^3}{2+x}$$

$2+x>0$

$$\frac{2x^3}{2+x}\leqslant\frac{2}{3}$$

$$6x^3\leqslant 4+2x$$

$$6x^3-2x-4\leqslant 0$$

$$2(3x^3-x-2)\leqslant 0$$

$$2(x-1)(3x^2+3x+2)\leqslant 0$$

	3	0	-1	-2
1		3	3	2
	3	3	2	0

$$x\leqslant 1 \ \wedge\ x\in(-2,2)$$

$$x\in(-2,1\rangle$$

D 40. Wykaż, że jeśli $q \in \left(-1, \frac{\sqrt{5}-1}{2}\right)$, to

$$\left(1+q+q^2+q^3+\ldots\right)\cdot\left[1-q(1+q)+q^2(1+q)^2-q^3(1+q)^3+\ldots\right]=\frac{1}{1-q^3}.$$

$1 + q + q^2 + q^3 + \ldots$

$a_1 = 1 \quad a_2 = q$

$q_1 = q$

$|q_1| < 1 \qquad q \in (-1, 1)$

$S_1 = \frac{1}{1-q}$

$1 - q(1+q) + q^2(1+q)^2 - q^3(1+q)^3 + \ldots$

$b_1 = 1 \quad b_2 = -q(1+q)$

$q_2 = -q(1+q)$

$|q_2| < 1$

$|-q(1+q)| < 1$

$|-q^2 - q| < 1$

$|q^2 + q| < 1$

$q^2 + q < 1 \quad \wedge \quad q^2 + q > -1$

$q^2 + q - 1 < 0 \qquad q^2 + q + 1 > 0$

$q_2 \in \left(\frac{-1-\sqrt{5}}{2}, \frac{-1+\sqrt{5}}{2}\right) \qquad q_2 \in R$

$q_2 \in \left(\frac{-1-\sqrt{5}}{2}, \frac{-1+\sqrt{5}}{2}\right)$

$S_2 = \frac{1}{1+q(q+1)}$

$q \in \left(\frac{-1-\sqrt{5}}{2}, \frac{-1+\sqrt{5}}{2}\right) \quad \wedge \quad q \in (-1, 1)$

$q \in \left(-1, \frac{-1+\sqrt{5}}{2}\right)$

$\left(1+q+q^2+q^3+\ldots\right)\cdot\left[1-q(1+q)+q^2(1+q)^2-q^3(1+q)^3+\ldots\right] =$

$= \frac{1}{1-q} \cdot \frac{1}{1+q(q+1)} = \frac{1}{(1-q)(1+q^2+q)} = \frac{1}{1+q^2+q-q-q^3-q^2} = \frac{1}{1-q^3}$ ▨